

UPDATES ON THE ML MODELLING TILL NOW:

Dataset Description

Each row in the dataset represents a **brand–influencer collaboration** and contains:

Key Categories of Attributes

1. Brand Information

- ACCOUNTNAME

2. Influencer Information

- INFLUENCER_ID
- NAME
- INSTAGRAM_USER_ID

3. Engagement Metrics

- INSTAGRAM_FOLLOWER_COUNT
- INSTAGRAM_TOTAL_ENGAGEMENT
- INSTAGRAM_TOTAL_LIKES
- INSTAGRAM_TOTAL_COMMENTS

4. Posting Activity

- INSTAGRAM_TOTAL_POSTS
- INSTAGRAM_POSTS_LAST_30_DAYS
- INSTAGRAM_POSTS_LAST_90_DAYS
- INSTAGRAM_POSTS_LAST_180_DAYS
- INSTAGRAM_POSTS_IN_LAST_1_YEAR

5. Campaign Interaction Metrics

- REFERRAL_LINK_TOTAL_CLICKS
- CAMPAIGN_OPT_INS

6. Target Variable

- TOTAL_SALES

Initial Approach

Model Used

- XGBoost Regressor

Baseline Performance

- R^2 Score: ~0.10–0.23
- RMSE: High
- MAE: High

Observation

- Model performance was weak due to:
 - Lack of strong predictive features
 - Presence of noisy data
 - Non-linear relationships in data

Trial and Error Process

4.1 Multiple Models Tested

Model	R ² Score Range	Observation
XGBoost	~0.10–0.25	Sensitive to feature quality
CatBoost	~0.17–0.30	Better with noisy data
Random Forest	~0.24	Most stable performance

Key Issues Identified

1. Data Leakage

- Features like:
 - TOTAL_ORDERS
 - AVG_ORDER_SIZE
 - TOTAL_COMMISSION
- These are derived from TOTAL_SALES
- Result: artificially high accuracy

2. Poor Feature Representation

- Raw counts were not sufficient
- Needed behavioral ratios

3. Outliers

- Large variations in follower counts and engagement

Feature Engineering Improvements

To improve model performance, additional derived features were created to better represent influencer behavior.

One important feature introduced was the Growth Rate (Posting Activity Index).

Growth Rate Formula:

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Growth Rate Formula:

$$\text{Growth Rate} = \frac{\left(\frac{\text{Posts}_{30}}{30}\right) + \left(\frac{\text{Posts}_{90}}{90}\right) + \left(\frac{\text{Posts}_{180}}{180}\right)}{\left(\frac{\text{Posts}_{365}}{365}\right)}$$

Interpretation:

- Measures recent posting activity relative to long-term posting behavior
- Value > 1 indicates increased recent activity
- Value ≈ 1 indicates consistent posting
- Value < 1 indicates declining activity

Purpose:

- Captures influencer consistency and momentum
- Helps model understand behavioral trends instead of relying only on raw counts

Impact:

- Provided a small improvement in model performance (~1–3% increase in R^2)
- Improved ranking stability of influencers

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Removal of Data Leakage

Removed post-outcome variables:

- TOTAL_ORDERS
- AVG_ORDER_SIZE
- TOTAL_COMMISSION
- NEW_CUSTOMERS
- PERCENTAGE_OF_BRAND_SALES

Result:

- Accuracy decreased but became realistic

Feature Engineering

Created meaningful features:

- engagement_rate
- click_through_rate
- engagement_per_post
- likes_per_follower
- comments_per_post
- posting_frequency_score

Purpose:

- Capture behavioral patterns instead of raw counts

Log Transformation

Applied log transformation to reduce skewness:

- followers_log
- engagement_log
- clicks_log

Why Accuracy is Limited

Despite improvements, accuracy remains moderate due to missing critical factors:

- Audience demographics

- Product relevance
- Brand strength
- Marketing strategy
- Content quality

These factors significantly influence sales but are not present in the dataset.

Why High Accuracy (90%+) is Not Achievable

Achieving R^2 of 0.9 requires:

- Complete representation of all influencing variables
- Minimal noise
- Direct causal features

In this dataset:

- Important variables are missing
- Relationships are complex and indirect

Therefore:

- High accuracy is not feasible
- Attempting to achieve it may lead to **overfitting or leakage**

Risk of Data Leakage

Definition

Data leakage occurs when the model uses information that would not be available at prediction time.

Examples in this project

- TOTAL_ORDERS
- AVG_ORDER_SIZE
- TOTAL_COMMISSION

Impact

- Artificially high accuracy
- Poor real-world performance

Mitigation

- Removed all post-outcome variables
- Ensured feature engineering used only pre-sales information

Model Input and Output

Model Input (Features)

The model takes the following types of inputs:

- Influencer metrics:
 - INSTAGRAM_FOLLOWER_COUNT
 - INSTAGRAM_TOTAL_ENGAGEMENT
 - INSTAGRAM_TOTAL_LIKES
 - INSTAGRAM_TOTAL_COMMENTS

- Activity metrics:
 - INSTAGRAM_POSTS_LAST_30_DAYS
 - INSTAGRAM_POSTS_LAST_90_DAYS
 - INSTAGRAM_POSTS_LAST_180_DAYS
- Campaign interaction metrics:
 - REFERRAL_LINK_TOTAL_CLICKS
- Engineered features:
 - engagement_rate
 - click_through_rate
 - engagement_per_post
 - likes_per_follower
 - comments_per_post
 - posting_frequency_score
 - log-transformed features

These features represent **influencer performance and behavior before or during collaboration**.

Model Output

The model outputs:

- **Predicted Total Sales (continuous value)** for each influencer
- This is then used to:
 - Rank influencers for a given brand
 - Identify top-performing influencers

Machine Learning Pipeline

The system follows a structured pipeline:

Step 1: Data Loading

- Load dataset containing brand–influencer collaborations
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Step 2: Data Cleaning

- Remove missing values
 - Replace infinite values
 - Drop leaky columns
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Step 3: Feature Engineering

- Create ratio-based features:
 - engagement_rate
 - click_through_rate
 - Create activity scores:
 - posting_frequency_score
 - Apply log transformations to reduce skewness
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Step 4: Feature Selection

- Remove:

- ID columns
 - Target column (TOTAL_SALES)
 - Retain only relevant predictive features
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Step 5: Train-Test Split

- Split dataset into:
 - Training set (80%)
 - Testing set (20%)
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Step 6: Outlier Handling

- Apply clipping using training data percentiles
 - Prevent extreme values from biasing the model
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Step 7: Model Training

- Train XGBoost / CatBoost / Random Forest models
 - Use hyperparameter tuning
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Step 8: Model Evaluation

- Metrics used:
 - R^2 Score

- RMSE
 - MAE
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Step 9: Cross-Validation

- Apply K-Fold cross-validation
 - Generate out-of-fold predictions
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Step 10: Prediction on Full Dataset

- Generate predictions for all rows
 - Store as PREDICTED_SALES
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Step 11: Brand-Based Filtering

- Input brand name
 - Filter dataset for that brand
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Step 12: Aggregation

- Group by influencer
 - Aggregate predicted sales
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Step 13: Ranking

- Sort influencers by predicted sales
- Output ranked list

Final System Workflow

1. Train model on historical collaboration data
2. Take brand name as input
3. Filter dataset for that brand
4. Predict sales for each influencer
5. Aggregate results per influencer
6. Rank influencers by predicted sales